



Applied Chemistry

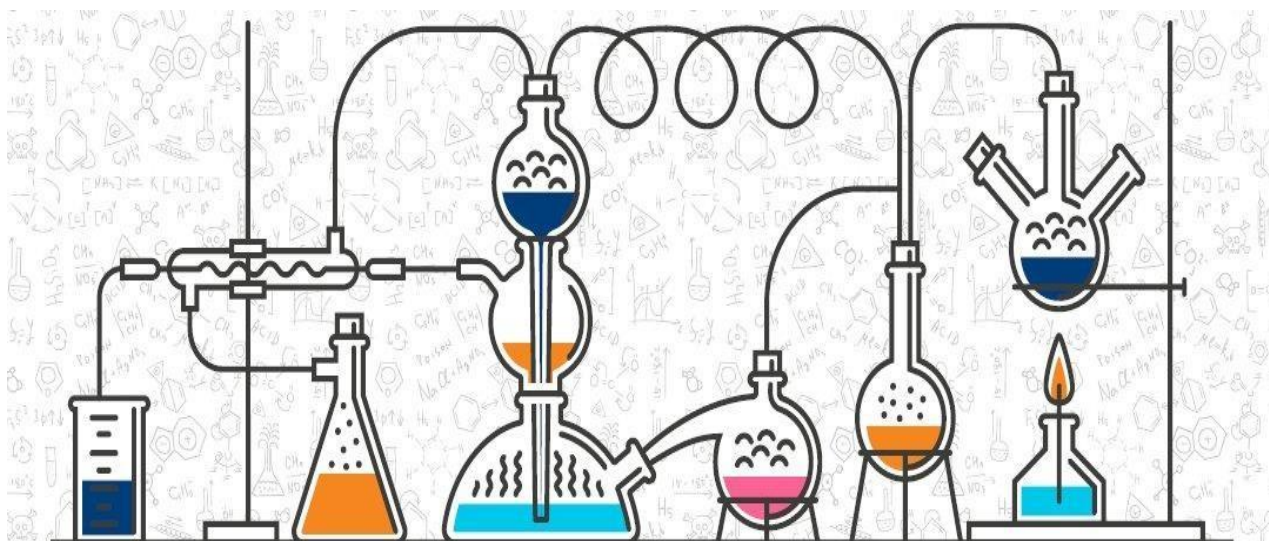
Chemistry of Smart Energy Systems- CS Stream

Stream: CS (Common to CS, IS, AI, CD, CY, CG, IC & BT Programs)

I / II SEMESTER

Course Code: 1BCHE102/202

Module - IV



DAYANANDA SAGAR COLLEGE OF ENGINEERING

Accredited by National Assessment & Accreditation Council (NAAC) with 'A' Grade
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9001:2015 Certified)

DEPARTMENT OF CHEMISTRY

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2025-26

Green Chemistry and Corrosion Science

Green Chemistry: Introduction, principles of green chemistry, properties and applications of green solvents for server heat management. Biosynthesis and properties of glycerol trioleate ester for server and IT infrastructure applications. Green synthesis of ZnO nanoparticles for magnetic Radio Frequency Identification (RFID) & Internet of Nano Things (IONT) system applications

Corrosion: Introduction, electrochemical theory of corrosion (Computer case as an example), types- differential metal and differential aeration corrosion (Waterline and pitting), corrosion control- Galvanization and anodization, vapour corrosion inhibitors for PCB's, corrosion penetration rate (CPR)- definition, importance and numerical problems.

GREEN CHEMISTRY:

Introduction to Green Chemistry

Green chemistry focuses on designing chemical products and processes that reduce or eliminate hazardous substances, promoting sustainability, environmental safety, and energy efficiency.

In early days, the extent of pollution was on a high end as various chemical industries released this whole lot of hazardous chemical waste into atmosphere. Hence the phenomenon of green chemistry was proposed to prevent the pollution caused by these chemical industries, which involves producing the same chemicals such as fertilizers, pesticides etc in lot more environment friendly manner/process. It is defined as the utilization of a set of principles that reduces or eliminates the use or generation of hazardous substances in the design, manufacture and application of chemical products. The term 'hazardous' is used in its broadest context which includes physical (e.g., explosive, flammable), toxicological (e.g., carcinogenic, mutagenic) and global (e.g., ozone depletion, climate change) factors.

Principles of Green Chemistry

There are 12 important principles of green chemistry. These principles help chemists and industries reduce pollution, save energy, and make safer chemicals.

1. Prevention of Waste

Chemical reactions should be designed to produce little or no waste and by-products. Less waste means less pollution and lower treatment cost.

2. Maximum Use of Reactants (Atom Economy)

All starting materials should be converted as much as possible into the final product. Very little material should be wasted during the reaction.

3. Avoid Hazardous Chemicals

Chemical processes should avoid producing toxic or harmful substances. This protects both human health and the environment.

4. Designing Safer Chemicals

Chemicals and products should be safe to use and should not cause harm to humans or nature.

5. Reduce Energy Consumption

Chemical reactions should use minimum energy. Reactions carried out at room temperature or with catalysts help save energy.

6. Use Safer Solvents

Solvents used in reactions should be environmentally friendly and non-toxic. Water or solvent-free reactions are preferred whenever possible.

7. Use Renewable Raw Materials

Starting materials should come from renewable sources such as plants, agricultural products, or natural gases rather than petroleum-based resources.

8. Minimize Use of Protecting Groups

Extra steps such as adding and removing protecting groups should be avoided because they produce additional waste and chemicals.

9. Use of Catalysts

Catalysts should be preferred because they increase reaction efficiency without being consumed in the reaction.

10. Design Biodegradable Products

Chemical products should break down naturally after use and should not remain in the environment for a long time.

11. Safer Manufacturing Processes

Chemical industries should be designed to prevent accidents, explosions, leaks, and harmful gas releases.

12. Improve Analytical Techniques

Analytical methods should use smaller amounts of chemicals and allow recycling of unused reagents whenever possible.

Introduction to Green Solvents:

Green solvents are environmentally benign liquids used to dissolve substances or facilitate chemical processes without harming humans or ecosystems. They aim to replace hazardous organic solvents with safer alternatives.

- **Water** – the ultimate green solvent.
- **Supercritical fluids** – like supercritical CO₂ for efficient heat transfer.
- **Ionic liquids (ILs)** – tunable, non-volatile, and thermally stable.
- **Bio-based solvents** – derived from renewable feedstocks like glycerol or ethyl lactate.

Properties of Green Solvents for Server Heat Management

For modern server cooling and data center thermal management, green solvents must have specialized properties:

1. High Thermal Conductivity & Heat Capacity – Efficiently absorb and transfer heat from CPUs, GPUs, and other components.
2. Low Viscosity – Ensures smooth circulation in cooling systems with minimal energy consumption.
3. Electrical Non-Conductivity – Prevents short circuits when in contact with electronics.
4. Chemical & Thermal Stability – Resistant to degradation at high temperatures; protects hardware from corrosion.
5. Non-Toxic & Biodegradable – Safe for humans and the environment, reducing risk in case of leaks.
6. Freeze & Boil Point Stability – Maintains performance under varying operating temperatures.
7. Leak Detection Capability – Some solvents are dyed or fluorescent for easy monitoring.

Examples: Propylene glycol-based fluids, silicone-based immersion fluids, and specially formulated ionic liquids. These are engineered to be compatible with metals like aluminum, copper, and steel in server hardware.

Green Solvents Applications in Server Heat Management:

Green solvents are revolutionizing data centre cooling through advanced technologies:

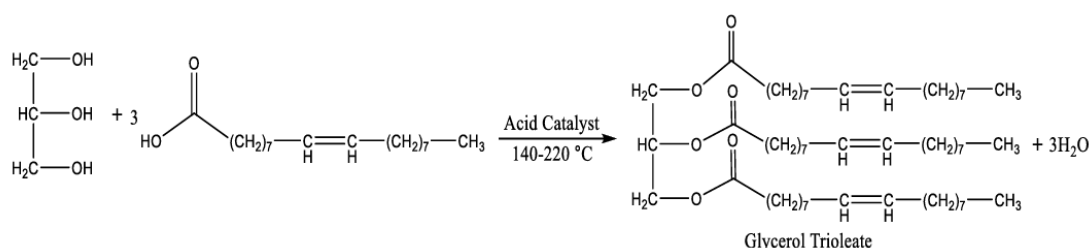
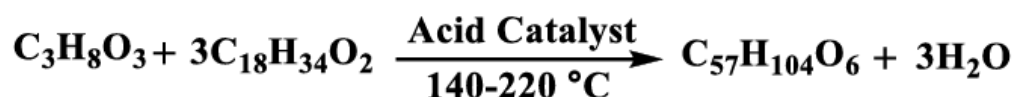
- **Direct Liquid Cooling (DLC):** Solvents circulate directly over CPUs, GPUs, and power components, removing heat efficiently and reducing energy consumption.
- **Immersion Cooling:** Entire servers are submerged in non-conductive green solvents, enabling higher server density, uniform cooling, and reduced infrastructure footprint.
- **Hybrid Cooling Systems:** Combines air and liquid cooling using green solvents to optimize thermal efficiency and energy use.
- **Sustainable Operations:** Green solvents with low global warming potential (GWP) and zero ozone depletion potential (ODP) help achieve eco-friendly, carbon-neutral data center operations.

Biosynthesis and properties of glycerol trioleate ester for server and IT infrastructure applications.

Synthesis of Glycerol Trioleate

On esterification reaction between glycerol (a triol) and oleic acid (a fatty acid) in presence of acid catalyst (e.g., *p*-toluenesulfonic acid or H₂SO₄) under heat 140–220°C to form Glycerol Trioleate.

Chemical Reaction:



Properties of glycerol trioleate:

CS Stream

Property	Value / Description
Physical state	Oily liquid at room temperature
Melting point	~-5.5 °C
Solubility	Insoluble in water, soluble in organic solvents
Major natural source	Olive oil (4–30%)
Biological role	Food, research, "Lorenzo's oil"
Combustion reaction	$C_{57}H_{104}O_6 + 80 O_2 \rightarrow 57 CO_2 + 52 H_2O$
Heat of combustion	8,389 kcal/mol (35,100 kJ/mol)

Uses in server & IT Infrastructure Applications:**Green Dielectric Coolant:**

Used as a non-toxic insulating liquid in data centre transformers and server cooling systems. Prevents short circuits and improves heat dissipation.

High-Temperature Lubricant for Hardware: Can act as a biodegradable lubricant for mechanical components like hard disk drives or server fans.

Phase Change Cooling Media: With high thermal stability, used in immersion cooling of CPUs and GPUs to maintain optimal temperatures. Reduces thermal hotspots in high performance computing.

Fire-Safe Alternative to Mineral Oils: Low flammability makes it a safer alternative for liquid cooling in server racks.

Eco-Friendly Thermal Management: Being biodegradable and non-toxic, it minimizes environmental hazards in IT facility maintenance.

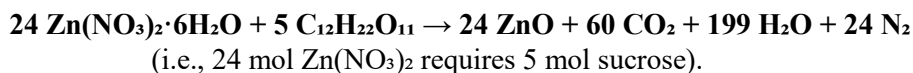
Enhanced Reliability of Electronics: Reduces thermal stress on sensitive ICs, prolonging the lifespan of critical infrastructure.

Green Solution Combustion Synthesis of ZnO Nanoparticles and its uses in magnetic Radio Frequency Identification (RFID) and Internet of Nano Things (IONT) system applications.

Solution combustion synthesis (SCS) is a rapid and energy-efficient route for producing zinc oxide (ZnO) nanoparticles. The process becomes greener when sugarcane juice is used as the fuel. Sugarcane juice contains sucrose, glucose, and fructose, which act as natural reducing and complexing agents. During combustion, these sugars decompose to release gases (CO₂, H₂O), creating porosity and leaving ZnO nanoparticles as the final product.

- Dissolve 3.65g of zinc nitrate hexahydrate [Zn(NO₃)₂·6H₂O] in distilled water.

$$Zn(NO_3)_2 \rightarrow Zn^{2+} + 2NO_3^-$$
- Add 7.3 mL Sugarcane juice typically has ~12% w/v sugar in the stoichiometric fuel-to oxidizer ratio. For sucrose (C₁₂H₂₂O₁₁) as the main fuel.
- Stir and ultrasonicate the solution for 30 minutes to form a homogeneous mixture.
- Heat gently at ~80–90 °C until a viscous gel forms.
- Place the gel in a preheated furnace (300–600 °C).
- Combustion occurs spontaneously, producing flame and froth as gases (CO₂, H₂O, N₂)



- Cool to obtain porous ZnO ash. Calcine the powder at 400–500 °C for 2–3 h to remove carbon residues and enhance crystallinity.

ZnO Nanoparticles in Magnetic RFID

- **Magnetic Properties:** Doping ZnO with Fe, Ni, or Co induces room-temperature ferromagnetism.
- **Enhanced Data Storage:** Magnetic ZnO allows higher data density and non-volatile memory in RFID tags.
- **Improved Security:** Magnetic behavior makes RFID tags harder to clone or tamper with.
- **Sensor Integration:** ZnO NPs can detect gases, chemicals, or temperature changes for smart RFID tags.
- **Chipless & Passive Tags:** ZnO/metal oxide nanocomposites (e.g., ZnO/TiO₂) enable passive sensing RFID systems.
- **Miniaturization & Flexibility:** Nanoparticle form allows small, flexible, and wearable RFID tags.

ZnO Nanoparticles in Internet of Nano Things (IoNT)

Nanoscale Sensing: High surface area and semiconducting properties enable sensitive detection of environmental or biomedical signals.

Energy Harvesting: Piezoelectric ZnO nanowires or NPs provide self-powered IoNT devices.

Flexible Electronics: ZnO NPs support durable, bendable nanosensors for health monitoring or smart infrastructure.

Targeted Biomedical Applications: Magnetic ZnO NPs can deliver drugs or monitor health internally.

Autonomous Networks: Magnetic interactions help self-assembly of nanosensor networks.

Cost-effective Large-scale IoNT: Enables distributed sensing systems for industrial, environmental, and medical applications.

Corrosion

Introduction: Corrosion is a process of destruction or deterioration of metal by the environment through chemical or electrochemical reactions. Most of the metals and alloys undergo corrosion except the noble metals. Corrosion is like “Cancer” for the metals. Eg: Rusting of Iron- it is due to formation of hydrated ferric oxide.

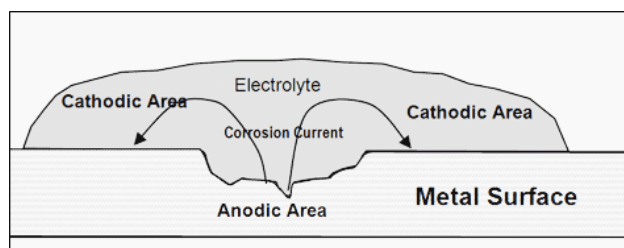
Electrochemical Theory of Corrosion (Computer case as an example)

Corrosion in computers is mainly an electrochemical process caused by moisture, oxygen, and environmental conditions. Computer cabinets are commonly made of steel containing iron, while internal parts contain metals such as copper and aluminum. During operation, computers heat up and cool down repeatedly, leading to condensation of moisture on metal surfaces. This thin moisture layer acts as an electrolyte and forms tiny electrochemical cells on the metal surface.

This type of corrosion is called wet or electrochemical corrosion, which occurs when metals are exposed to a conducting medium such as moisture. Corrosion may occur due to the formation of anodic and cathodic regions on the same metal surface or when two different metals are in contact in the presence of an electrolyte.

Effects of corrosion in computers include:

- Weakening of metal cabinets
- Increased electrical resistance
- Damage to connectors and ports
- Reduced grounding efficiency
- Failure of electronic components in severe cases



- Corrosion occurs due to the formation of anodic and cathodic regions on the same or different metals in a conducting medium.
- At the anode, oxidation takes place, and the metal corrodes into ions by releasing electrons.
- At the cathode, reduction occurs by consuming the liberated electrons, keeping the cathodic region protected.
- Corrosion continues as long as anodic and cathodic reactions proceed simultaneously, influenced by impurities and oxygen concentration differences.

Anodic Region-Oxidation

Cathodic region- Reduction

Corrosion current-flow of electron from anode to cathode

Corrosion product deposit between anode and cathode

At the Anodic Region:

Iron liberates Fe^{2+} ions and electrons at the anodic region:



At the Cathodic Region:

a. Liberation of hydrogen takes place in the absence of oxygen at the cathode surface.

- (i) In Acidic medium- $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
 (ii) In neutral/alkaline medium- $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{OH}^- + \text{H}_2$

b. Absorption of oxygen takes place in the presence of oxygen at the cathode surface.

- (i) In Acidic medium- $2\text{H}^+ + \frac{1}{2} \text{O}_2 + 2\text{e}^- \rightarrow \text{H}_2\text{O}$
 (ii) In neutral/alkaline medium - $\text{H}_2\text{O} + \frac{1}{2} \text{O}_2 + 2\text{e}^- \rightarrow 2\text{OH}^-$

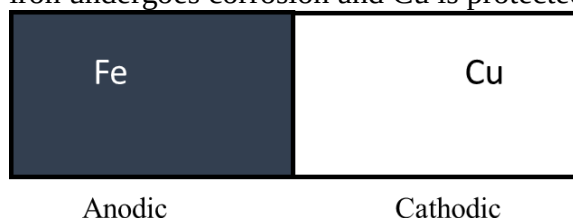
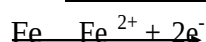
Formation of Rust:

- (i) In the presence of excess oxygen
 $\text{Fe}^{2+} + 2\text{OH}^- \rightarrow \text{Fe}(\text{OH})_2$
 (ii) In the presence of limited oxygen
 $4\text{Fe}(\text{OH})_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2[\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}]$ (Brown rust)

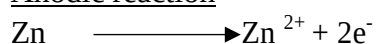
Types of corrosion:**Differential Metal Corrosion** (galvanic corrosion):

When two dissimilar metals are in contact with each other in a corrosive conducting medium. The two metals differ in their tendencies to undergo oxidation; the one with lower electrode potential or the more active metal acts as anode and the one with higher electrode potential or the less active metal act as cathode the potential difference between the two metals is the cause corrosion. The anodic metal undergoes corrosion and the cathodic metal is protected from corrosion.

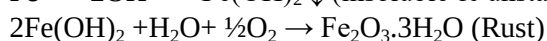
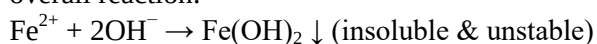
When two metals Fe and Cu are in contact: Since iron has lower reduction potential compared to Cu, iron undergoes corrosion and Cu is protected.

Anodic reactionCathodic reaction

When two metals Zn and Fe are in contact: Since Zinc has lower reduction potential compared to Fe, Zn undergoes corrosion and Fe is protected.

Anodic reactionCathodic reaction

overall reaction:

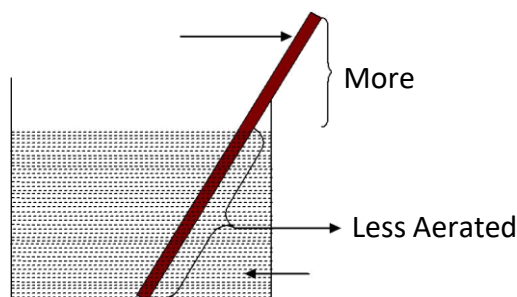
**Other examples:**

1. Steel screws in copper sheet
2. Steel screws with copper washer
3. Nut and bolts are made of different metals.

Differential aeration Corrosion

When the metal surface is exposed to a varying aeration, one part of the metal poorly oxygenated become anode and corroded. Another part is more oxygenated acts as cathode and protected. This type of corrosion is known as differential aeration corrosion.

Differential aeration corrosion further classified into two types.



Reactions

At anode : $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$ (oxidation)

At cathode : $\frac{1}{2}\text{O}_2 + \text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{OH}^-$ (Reduction)

$\text{Fe}^{2+} + 2\text{OH}^- \rightarrow \text{Fe}(\text{OH})_2$

$2\text{Fe}(\text{OH})_2 + \frac{1}{2}\text{O}_2 + \text{H}_2\text{O} \rightarrow \text{Fe}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow \text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}(\text{rust})$

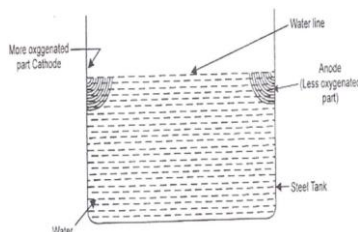
Example

- Part of nail inside the wall undergoes corrosion.
- Partially buried pipeline in soil undergoes corrosion.
- Partially filled iron tank undergoes corrosion inside water.

This differential aeration corrosion is divided into two types

a. Water-line corrosion

When water is stored in a metallic tank, the area below the water level is poorly oxygenated and acts as anode, whereas area above the water level is highly oxygenated and acts as cathode. The

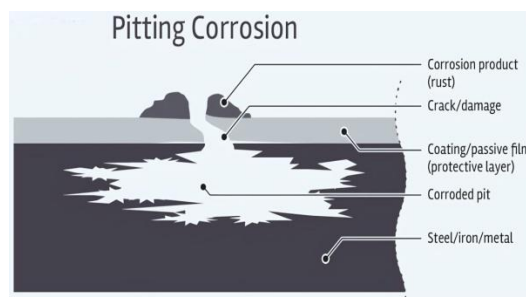


anode is corroded, whereas cathode is remains protected from corrosion. Water line corrosion is a problem for the marine engineers, when marine plants are attached to the ships in ocean. This type of corrosion is commonly observed in ships floating in seawater for long periods.

Fig: Water line corrosion

2. Pitting corrosion

It occurs when small particles of dust, water drop on metal surface or breakdown of protective film. A typical pitting corrosion is illustrated as shown in figure. The metal portion covered by dust act as anode due to less oxygenated and undergoes corrosion beneath the dust forming a pit. The metal part free from dust exposed to higher oxygen concentration become cathode and protected from corrosion. The corrosion rate is higher in this case due to small anode area and large cathode area. All the electrons liberated at anode (pit) are absorbed by large cathode area for reduction reaction.



At anode: $\text{Fe} \rightarrow \text{Fe}^{2+} + 2e^-$

At cathode: $\text{H}_2\text{O} + \frac{1}{2}\text{O}_2 + 2e^- \rightarrow 2\text{OH}^-$

overall reaction:

$\text{Fe}^{2+} + 2\text{OH}^- \rightarrow \text{Fe}(\text{OH})_2 \downarrow$ (insoluble & unstable)

$2\text{Fe}(\text{OH})_2 + \text{H}_2\text{O} + \frac{1}{2}\text{O}_2 \rightarrow \text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$ (Rust)

Corrosion control:

According to electrochemical theory, corrosion occurs due to the oxidation and reductions reactions at the anodic and cathodic areas. These reactions occur simultaneously and are interdependent. So by decreasing the rate of any of these reactions or by preventing any one reaction, the rate of corrosion can be decreased. The methods used to control corrosion are:

1. Metal Coating
 - a) Anodic Metal Coating: Galvanisation
 - b) Cathodic Metal Coating: Tinning
2. Cathodic protection- Sacrificial Anode Method
3. Anodization
4. Corrosion inhibitors

1. Metal Coating

a. Metal Coatings

The coating metal deposited on base metal is known as metal coating. Metal coatings are done by hot dipping, electroplating, electroless plating, thermal spraying etc.

The metal coating is classified into two types 1. Anodic coatings 2. Cathodic coatings.

Anodic coatings.

Base metal coated with anodic and active metals, which are anodic to base metal. For example, iron is coated with active metals like Mg, Zn, Al etc.

Galvanization is an example for anodic coating. It is extensively used to protect iron and steel materials. It is done by hot dipping.

The following steps are involved in the galvanization process

- 1) The iron sheet is washed with solvent to remove grease & oily materials (degreasing).
- 2) The iron sheet is washed with $\text{dil. H}_2\text{SO}_4$ to remove oxide film (descaling) and rinsed with distilled water.
- 3) The sheet is then treated with a mixture of aqueous solution of zinc chloride and ammonium chloride & dried.
- 4) The dried sheet is dipped in molten zinc at $430 - 450^\circ\text{C}$
- 5) Excess Zn is removed by rolling, wiping etc.

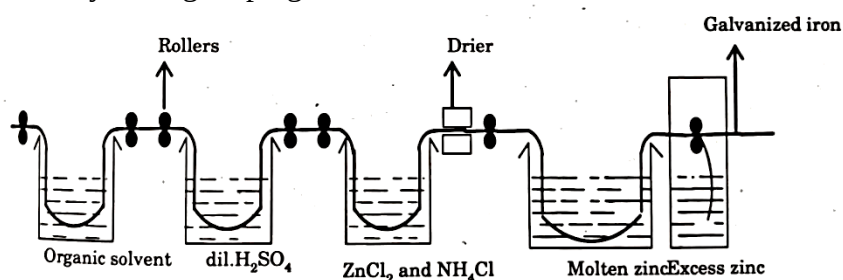


Figure-Galvanization process**b. Anodization- Anodized coating of Aluminium.**

Anodized coating is generally produced on non-ferrous metals like Al, Zn, Mg made as anode, Pb is generally used as cathode in an electrolyte by passing direct current. The most commonly used electrolytes are H_2CrO_4 , H_2SO_4 , H_3PO_4 , $(\text{COOH})_2$, H_3BO_3 or their mixtures.

The anodic metal oxide is porous and provides good adherence for paints and dyes. The strength and corrosion resistance of the anodized film can be increased by sealing, which involves heating in boiling water or steam or metal salt solutions. The treatment changes porous alumina at the surface of coating into its monohydrate ($\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$), which occupies more volume, thereby the pores are sealed.

Anodized aluminium can be produced in a number of coloured finishes.

1. An organic dye is adsorbed on the freshly anodized surface and it is then sealed.
2. A metal Ni, Co, Sn is deposited into the porous, resulting electro colouring process.

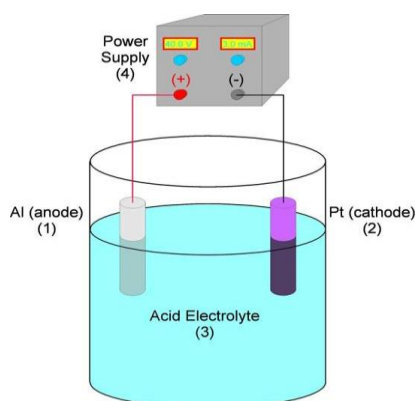
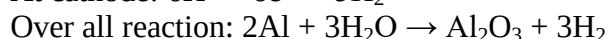
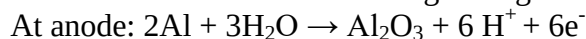


Figure-Electrolytic cell for anodizing aluminium.

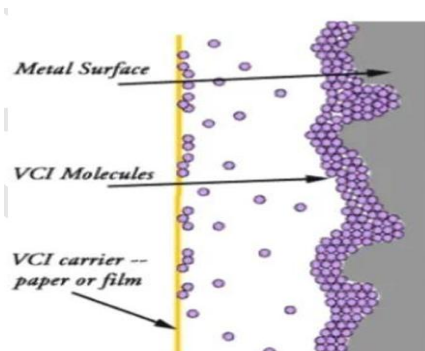
The electrode reactions occurring during anodising process are:



Uses: It is used as soap boxes, tiffin boxes, window frames etc. Anodic films are also used for number of cosmetic effects.

Vapour Corrosion Inhibitors (VCIs) for Protecting Computer Circuit Boards:

Vapor corrosion inhibitors (VCIs) are chemical compounds that protect metals from corrosion by releasing vapor that forms a protective layer on metal surfaces. VCIs are often used in situations where it is impractical or undesired to use rust-preventative liquids or other surface treatments.

**Principle:**

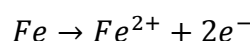
- The compounds that VCIs release into an airspace actively prevent corrosion by forming strong bonds with a metal surface, promoting and maintaining the natural and healthy passive oxide layer on the metal.
- The VCI layer that form is invisible, dry, and does not affect the physical properties or functionality of the metal in any way. The protected metal can be painted, treated, and/or used straight out further cleaning.
- The inhibitor is transmitted as vapours, and the vapour phase is controlled by the structure of the crystal lattice and the character of atomic bond in the molecule.
- The protective vapours expand within the enclosed space until the equilibrium, which is determined by their partial pressure.

Mechanism by keeping Iron (Fe) or Steel Protection by VCI

Dicyclohexyl ammonium nitrite (DICHAN) is a commonly used Vapour Corrosion Inhibitor (VCI) for protecting iron and steel.

- DICHAN slowly releases protective vapours into the surrounding air.
- These vapours reach the iron or steel surface and condense to form a thin invisible layer.
- In the presence of moisture, the inhibitor dissolves slightly and releases nitrite ions.
- The nitrite ions form a protective passive oxide layer on the steel surface, preventing rust formation.
- This protective layer blocks the contact of oxygen and moisture with the metal, thereby reducing electrochemical corrosion.

The oxidation reaction on iron surface is:



Because of the protective VCI layer, the movement of iron ions and electrons is reduced, thereby slowing down corrosion and rust formation.

Corrosion penetration rate (CPR):

Corrosion penetration rate is defined in three ways

- The speed at which any metal in a specific environment deteriorates due to chemical reaction in the metal when it is exposed to a corrosive environment.
- The amount of corrosion loss per year in thickness.
- The speed at which corrosion spreads to the inner portions of a material. The speed or rate of deterioration depends on the environment conditions and the type of metal under study.

The equation for CPR (Corrosion Penetration Rate) is:

$$CPR = \frac{KW}{\rho At}$$

Where: CPR = Corrosion Penetration Rate (measured in miles per year, mpy)

1 mpy = 0.001 inch per year

K = A constant

W = Total weight loss due to exposure to a corrosive environment

ρ = Density of the material

A = Surface area of the exposed metal

t = Time taken for the loss of metal (exposure time)

Constant	CPR in mpy	CPR mmpy
K	534	87.6
W (wt loss)	mg	Mg

ρ	g/cm^3	g/cm^3
A	inch^2	cm^2
t	hrs	Hrs

Conversions:1 Gram = 1000 Milligram or 10^3 1 Kg = 1000000 Milligram or 10^6

1 Day = 24 Hours

1 Month = 30 days = 30×24 Hours1 Year = $1 \times 365 \times 24$ Hours

1 Square Centimeter = 0.155 square inch

1 square inch = 6.4516 sq cm

Importance of CPR

- Predicts Service Life of Materials**

CPR helps estimate how long a metal component (e.g., pipelines, storage tanks, structural frames, computer cabinets) will last before failure due to thinning.

- Material Selection**

By comparing CPR values, engineers can select suitable corrosion-resistant materials for specific environments.

- Design Safety**

CPR data helps determine the required thickness of metal components to ensure long-term durability.

- Maintenance Planning**

It allows scheduling of inspections, repainting, or replacement before serious damage occurs.

- Cost Reduction**

Preventive action based on CPR minimizes unexpected breakdowns and reduces repair costs.

- Evaluation of Protective Methods**

CPR is used to assess the effectiveness of coatings, inhibitors, cathodic protection, and galvanization.

- Industrial Safety**

In industries such as oil & gas, chemical plants, and power stations, CPR helps prevent catastrophic failures due to corrosion.

1. A thickness of brass sheet of area 400 inch^2 is exposed to air near the ocean. After two years period it was found to experience a weight loss of 375g due to corrosion. If the density of Brass is 8.73 g/cm^3 . Calculate the CPR in mm/year mpy.

Solution:Given Area $A = 400 \text{ in}^2$ Weight loss $W = 375 \text{ g} = 375,000 \text{ mg}$ Density of Brass $\rho = 8.73 \text{ g/cm}^3$ Time $t = 2 \text{ years} = 2 \times 365 \times 24 = 17,520 \text{ hours}$

$$CPR = \frac{KW}{\rho At}$$

CPR in mpy

$K = 534$ $W = 375000 \text{ mg}$ $\rho = 8.73 \text{ g/cm}^3$ $A = 400 \text{ in}^2$ $t = 17,520 \text{ hrs}$	$CPR \text{ mpy} = \frac{534 \times 375000}{8.73 \times 400 \times 17520}$ $CPR \text{ mpy} = \frac{200250000}{61186560}$ $\approx 3.27 \text{ mpy}$
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CPR in mm/year (mmpy)

$K = 87.6$ Convert area from in^2 to cm^2 $A = 400 \text{ in}^2 = 400 \times 6.4516 = 2580.64 \text{ cm}^2$ $W = 375000 \text{ mg}$	$CPR \text{ mmpy} = \frac{87.6 \times 375000}{8.73 \times 2580.64 \times 17520}$ $CPR \text{ mmpy} = \frac{32850000}{394248859.4}$ $\approx 0.0833 \text{ mm/year}$
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$\rho = 8.73 \text{ g/cm}^3$ $t = 17,520 \text{ hrs}$	
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- 2) An iron metal lost $2.4 \times 10^{-3} \text{ g}$ of weight when it is immersed in deaerated acidic solution for 2 hrs. Calculate the corrosion penetration rate (CPR) for the exposed area of 4 cm^2 . Given $k = 534$ and the density of metal $= 5.37 \text{ g cm}^{-3}$.

Solution:

$$\begin{aligned}
 CPR &= \frac{KW}{\rho At} \\
 &= \frac{534 \times 2.4 \times 10^{-3}}{5.37 \times 4 \times 2} \\
 &= 0.027 \text{ cm/hr}
 \end{aligned}$$

- 3) An iron metal lost $3.3 \times 10^{-3} \text{ g}$ of weight when immersed in deaerated acid solution for 3 hour. Calculate the corrosion penetration rate for the exposed area of 5 cm^2 . Given $K=534$ and density of metal $= 5.37 \text{ g cm}^{-3}$.

$$\begin{aligned}
 CPR &= \frac{KW}{\rho At} \\
 &= \frac{534 \times 3.3 \times 10^{-3}}{5.37 \times 5 \times 3} \\
 &= 0.022 \text{ cm/hr}
 \end{aligned}$$

- 4) A sheet of carbon steel one meter wide by three meters long has lost 40g to corrosion over the past six months. Convert that mass loss to a penetration rate of the steel in mm units and mpy units. What could be the corrosion rate? (Carbon steel density $= 7.8 \text{ g/cm}^3$) To calculate CPR in mmpy. Can we apply such steel for construction purpose?

Solution: To calculate CPR in mmpy Given;

$$K = 87.6$$

$$W (\text{wt loss}) = 40 \text{ g} = 40 \times 1000 \text{ mg} \quad \rho = 7.8 \text{ g/cm}^3 = 7.8 \text{ g/cm}^3$$

$$A = 1 \text{ m} \times 3 \text{ m} = 3 \text{ m}^2 = 3 \times 100 \times 100 \text{ cm}^2 \quad T = 6 \text{ months} = 6 \times 30 \times 24 \text{ hrs}$$

We have

$$CPR = \frac{KW}{\rho At} = \frac{87.6 \times 40 \times 1000}{7.8 \times 3 \times 100 \times 100 \times 6 \times 30 \times 24}$$

$$\text{CPR} = 3.466 \times 10^{-3} \text{ mmpy}$$

To calculate CPR in mpy Given;

$$K = 534$$

$$W (\text{wt loss}) = 40 \text{ g} = 40 \times 1000 \text{ mg} \quad \rho = 7.8 \text{ g/cm}^3 = 7.8 \text{ g/cm}^3$$

$$A = 1 \text{ m} \times 3 \text{ m} = 3 \text{ m}^2 = 3 \times 1550 \text{ inch}^2 \quad T = 6 \text{ months} = 6 \times 30 \times 24 \text{ hrs}$$

$$\text{CPR} = \frac{534 \times 40 \times 1000}{7.8 \times 3 \times 1550 \times 6 \times 30 \times 24}$$

$$= 0.0136 \text{ mpy}$$

2. An iron metal lost $2.4 \times 10^{-3} \text{ g}$ of weight when it is immersed in deaerated acidic solution for 2 hrs. Calculate the corrosion penetration rate (CPR) for the exposed area of 4 cm^2 . Given $k = 534$ and the density of metal $= 5.37 \text{ g cm}^{-3}$.

Solution:

$$\text{Corrosion penetration rate (CPR)} = kW / \rho At$$

$$= \frac{534 \times 2.4 \times 10^{-3}}{5.37 \times 4 \times 2}$$

$$5.37 \times 4 \times 2$$

$$= 0.027 \text{ cm/hr}$$

2. An iron metal lost $3.3 \times 10^{-3} \text{ g}$ of weight when immersed in deaerated acid solution for 3 hour. Calculate the corrosion penetration rate for the exposed area of 5 cm^2 . Given $K=534$ and density of metal $= 5.37 \text{ g cm}^{-3}$.

$$\text{Corrosion penetration rate (CPR)} = kW / \rho At$$

$$= \frac{534 \times 3.3 \times 10^{-3}}{5.37 \times 5 \times 3}$$

$$5.37 \times 5 \times 3$$

$$= 0.022 \text{ cm/hr}$$

3. Calculate the CPR in both mpy and mmpy for a thick steel sheet of area 100 inch^2 which experiences a weight loss of 485 g after one year. (Density of steel $= 7.9 \text{ g/cm}^3$).

Solution: To calculate CPR in mmpy

$$\text{Given; } K = 87.6$$

$$W (\text{wt loss}) = 485 \text{ g} = 485 \times 1000 \text{ mg}$$

$$\rho = 7.9 \text{ g/cm}^3 = 7.9 \text{ g/cm}^3$$

$$A = 100 \text{ inch}^2 = 100 \times 6.45 \text{ cm}^2$$

$$t = 1 \text{ year} = 365 \times 24 \text{ hrs}$$

$$1 \text{ inch}^2 = 6.45 \text{ cm}^2$$

$$1 \text{ cm}^2 = 0.155 \text{ inch}^2$$

$$\text{We have, Corrosion penetration rate (CPR)} = kW / \rho At$$

$$CPR = \frac{87.6 \times 485 \times 1000}{7.9 \times 100 \times 6.45 \times 365 \times 24}$$

$$CPR = 0.9518 \text{ mmpy}$$

To calculate CPR in mpy

Given; K = 534

$$W (\text{wt loss}) = 485 \text{ g} = 485 \times 1000 \text{ mg}$$

$$\rho = 7.9 \text{ g/cm}^3 = 7.9 \text{ g/cm}^3$$

$$A = 100 \text{ inch}^2 = 100 \text{ inch}^2$$

$$t = 1 \text{ year} = 365 \times 24 \text{ hrs}$$

We have, **Corrosion penetration rate (CPR) = kW/ ρAt**

$$CPR = \frac{534 \times 485 \times 1000}{7.9 \times 100 \times 365 \times 24}$$

$$CPR = 37.424 \text{ mpy}$$

1. Write the principles of green chemistry.
2. Properties of green solvents for server heat management
3. Applications of green solvents in server heat management.
4. Biosynthesis and properties of triolate ester for server and its infrastructure applications.
5. Green synthesis of ZnO nanoparticles and its uses in RFID and IONT applications.
6. Write a note on electrochemical theory of corrosion by taking computer case as an example
7. Write a note types of corrosion.
8. What's is aeration corrosion and explain its types with neat diagram
9. Explain the process of galvanization with labelled diagram
10. Explain the process of anodization.
11. Describe the principle and mechanism of vapour corrosion inhibitors
12. Explain the importance of CPR.